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0178/03

May/June 2018

2 hours

Candidates answer on the Question Paper.

Additional Materials: Electronic Calculator
Geometrical Instruments
Tracing Paper (optional)

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

Answer all questions.

If working is needed for any question it must be shown below that question.

- Electronic calculators should be used.

If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For π use either your calculator value or 3.142.

The number of marks is given in brackets [] at the end of each question or part question.
The total of the marks for this paper is 100.



This document consists of 16 printed pages.



Examinations Council of Lesotho

$$1 \quad \mathbf{A} = \begin{pmatrix} -1 & 2 \\ 0 & 1 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 2 & 0 \\ 1 & 4 \end{pmatrix}$$

Find the following matrices.

(a) $3\mathbf{A}$

Answer (a) [1]

(b) $\mathbf{A} + \mathbf{B}$

Answer (b) [2]

- 2 (a) Pule and Thabo receive a sum of money from their father, and share it in the ratio 3:2.
If Thabo received M 1430.40.

(i) Calculate the total amount received by the two boys.

Answer (a)(i) M [

- (ii) Thabo spends 65% of M 1430.40 to buy clothes.

Calculate the amount he spends on clothes.

Answer (a)(ii) M



(b) Manka invests M6000 in company A and M5000 in company B.

- Company A pays **simple** interest at the rate of 4% per year.
- Company B pays **compound** interest at the rate of 5% per year.

From which company will he get more interest after 3 years and by how much?

Answer (b) Company by M [5]



3 (a) $\vec{AB} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}, \vec{BC} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$

Find

(i) $2\vec{BC}$,

Answer (a)(i) [1]

(ii) $\vec{AC}$,

Answer (a)(ii) [1]

(iii) $|\vec{BC}|$

Answer (a)(iii) [2]

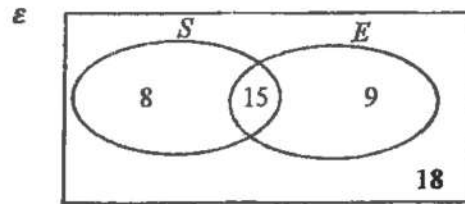
(b) A translation maps a point (3, 2) onto the point (4, 6).

Find the translation vector.

Answer (b) [2]



- 4 In a survey, 50 people are asked if their spoken language is Sesotho(S), English (E) or both. The information in the Venn diagram represents the number of people.



(a) Find

- (i) the total number of people who can speak **only one** language,

Answer (a)(i) [1]

- (ii) the number of people who can speak either Sesotho or English.

Answer (a)(ii) [1]

(b) Find

- (i) $n(S \cap E)$,

Answer (b)(i) [1]

- (ii) $n(S \cup E)$.

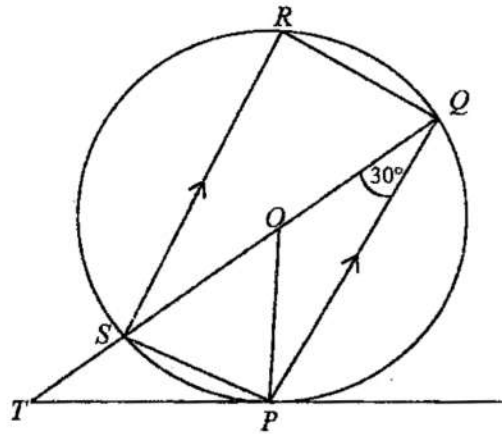
Answer (b)(ii) [1]

- (c) Use set notation to represent the number of people who can speak Sesotho only.

Answer (c) [2]



- 5 In the diagram, P , Q , R and S are points on the circle with centre O .



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Angle $PQQ = 30^\circ$ and TOQ is a straight line.

PQ is parallel to SR and TP is a tangent to the circle at P .

- (a) State the size of $\widehat{QSR}$ and give a reason for your answer.

Answer (a) $\widehat{QSR} = \dots\dots\dots$

Reason $\dots\dots\dots$ [2]

(b) Calculate

- (i) $\widehat{SPO}$,

Answer (b)(i) $\dots\dots\dots$ [1]

- (ii) $\widehat{TSP}$.

Answer (b)(ii) $\dots\dots\dots$ [1]



- 6 (a) Write as a single fraction.

$$\frac{x+2}{3} - \frac{2x-1}{2}$$

Answer (a) [3]

(b) $t = \frac{n+5}{n}$

- (i) Find t when $n = 3$.

Answer (b)(i) t [1]

- (ii) Make n the subject of the formula $t = \frac{n+5}{n}$.

Answer (b)(ii) n [3]

- (c) Factorise completely.

$$3a - 6b + ca - 2cb$$

Answer (c) [2]



7 (a) Solve the equations.

(i) $3x - 5 = 7$

Answer (a)(i) $x = \dots\dots\dots$ [2]

(ii) $\frac{3x-5}{2} = \frac{x+3}{4}$

Answer (a)(ii) $x = \dots\dots\dots$ [3]

(iii) $x^2 - 2x - 15 = 0$

Answer (a)(iii) $x = \dots\dots\dots$ or $\dots\dots\dots$ [3]

(b) Solve the simultaneous equations.

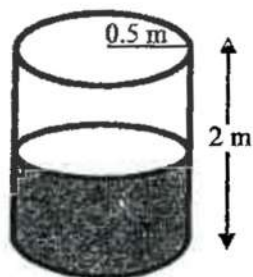
$$5x - 2y = 13$$

$$2x - 3y = 3$$

Answer (b) $x = \dots\dots\dots$ $y = \dots\dots\dots$ [3]



- 8 The diagram shows a cylindrical container of radius 0.5 m and height 2 m. The container is exactly half filled with water.



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- (a) Calculate the surface area of the (inside of the) container that is in contact with the water.

Answer (a) m^2 [3]

- (b) The container is now completely filled with water.

Calculate the **total volume**, in litres, of the water in the container.

Note ($1\text{m}^3 = 1000$ litres)

Answer (b) l [3]

- (c) The water is drawn from the container at the constant rate of 10 litres per minute.

How long will it take to draw 300 litres of water from the container?

Answer (c) minutes [2]

- 9 The table shows marks scored by 40 pupils in a Music class.

Marks	10	11	12	13	14	15
Frequency	5	5	7	4	10	9

(a) Find

(i) the mode,

Answer (a)(i) [1]

(ii) the median,

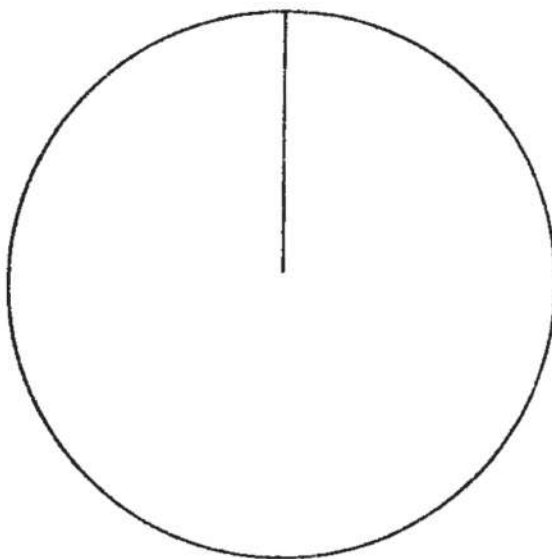
Answer (a)(ii) [2]

(iii) the mean.

Answer (a)(iii) [3]



- (b) Draw part of a pie chart to represent the number of students who scored 15 marks.



[3]

- (c) A pupil is chosen at random from the class.

Find the probability that the pupil scored

- (i) 8 marks,

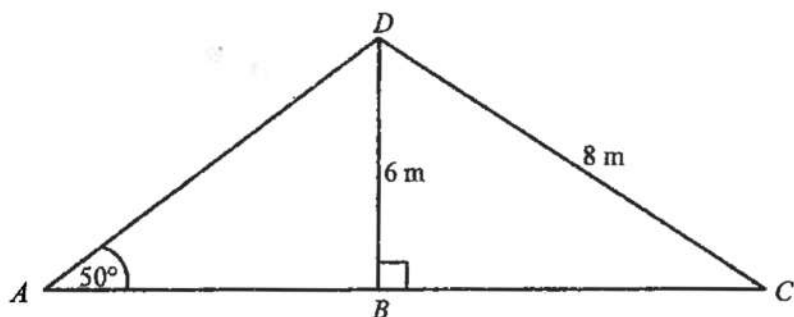
Answer (c)(i) [1]

- (ii) at least 13 marks.

Answer (c)(ii) [2]



- 10 The diagram shows triangle ACD and line BD which is perpendicular to AC .



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$BD = 6$ m, $CD = 8$ m and angle $BAD = 50^\circ$.

Calculate

(a) BC ,

Answer (a) m [2]

(b) angle DCB ,

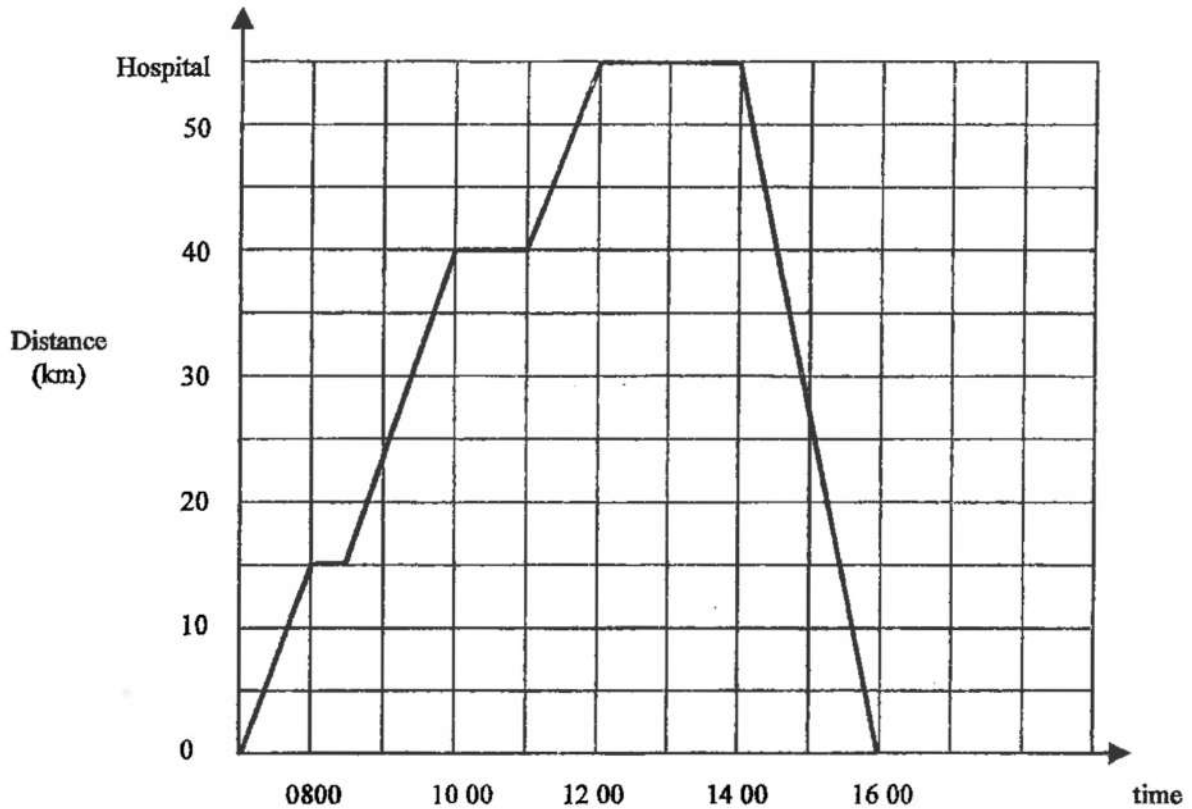
Answer (b) $\widehat{DCB}$ $^\circ$ [2]

(c) AD .

Answer (c) m [3]



- 11 The travel graph shows Mpho's journey to and from a hospital.



- (a) (i) Describe Mpho's motion at 08 00 hours.

Answer (a)(i) [1]

- (ii) Find the time at which he arrives at the hospital.

Answer (a)(ii) [1]

- (b) For how long does he stay at the hospital?

Answer (b) b [1]

- (c) Calculate Mpho's average speed, in km/h, on his way to the hospital.

Answer (c) km/h [3]



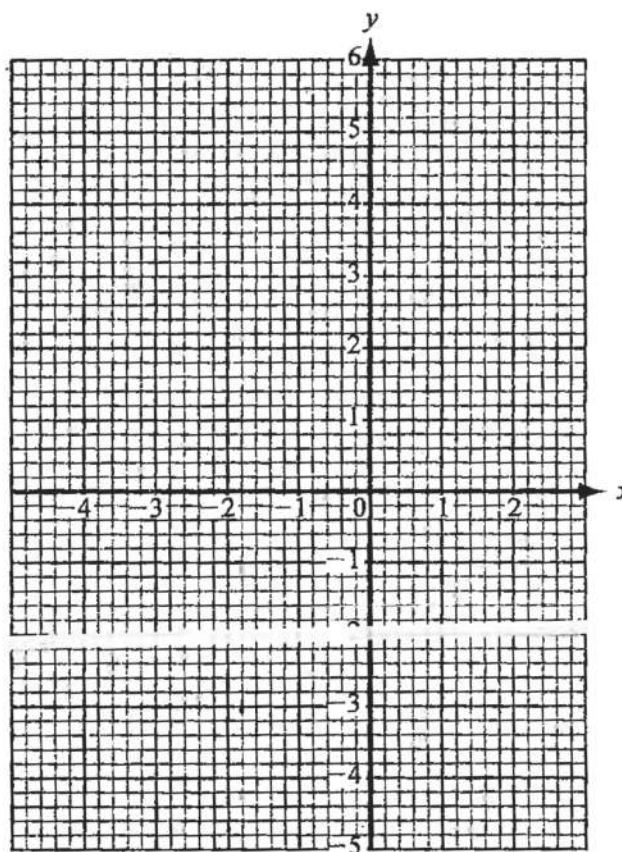
12 It is given that $y = x^2 + 2x - 3$.

(a) Complete the table for the values of y

x	-4	-3	-2	-1	0	1	2
y	5			-4	-3	0	

[2]

(b) On the grid draw the graph of $y = x^2 + 2x - 3$ for $-4 \leq x \leq 2$.



[3]

(c) Draw the line of symmetry of the graph.

[1]

(d) Use your graph to solve the equation $x^2 + 2x - 3 = 3$.

Answer (d) $x = \dots\dots\dots$ or $\dots\dots\dots$ [2]



- (e) On the same grid draw the graph of $y = 2x + 1$.

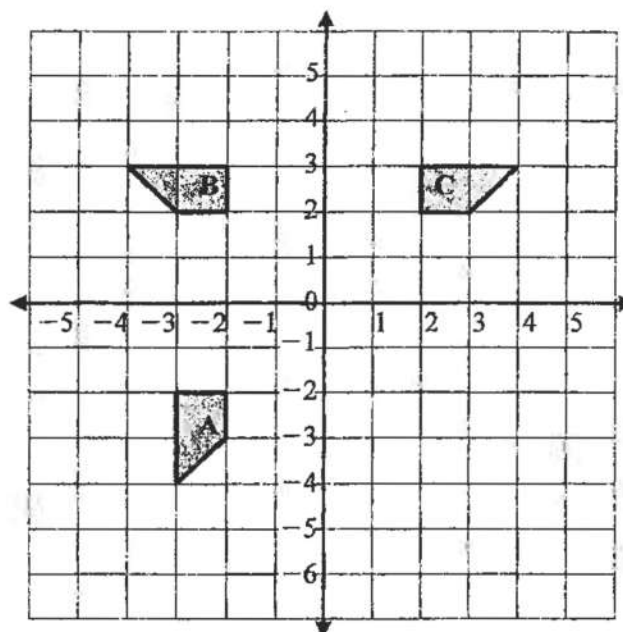
[2]

- (f) Write down the x -coordinates of the points where the graphs meet.

Answer (f) [1]



13 The diagram shows three trapeziums A , B and C .



(a) Describe fully the single transformation that maps

(i) A onto B

Answer (a)(i) [3]

(ii) A onto C .

Answer (a)(ii) [2]

(b) Enlarge shape B by scale factor -2 and centre $(-1, 1)$.

Label the image B_2 .

[3]

